

PROJECT WORK BREAKDOWN STRUCTURE (WBS)

CSE241: Restaurant Table Reservation & Ordering System

1. Project Overview & Architecture

This Work Breakdown Structure (WBS) outlines all deliverables, task packages, dependencies, and timeline estimates for the Desktop Restaurant Table Reservation & Ordering System. The project is split into two major milestones designed to build a robust backend followed by a feature-rich JavaFX interface with concurrency and socket networking.

2. Detailed Work Breakdown Schedule

WBS ID	Task Name	Phase	Duration	Start	End	Predecessors
1.1	Milestone 1: Core Backend & OOP Foundations	Phase 1	10 Days	2026-07-30	2026-08-08	-
1.1.1	Base Architecture & Shared Resources	Phase 1	2 Days	2026-07-30	2026-07-31	None
1.1.2	Core Domain Entities & Interfaces	Phase 1	3 Days	2026-08-01	2026-08-03	1.1.1
1.1.3	Data Store & Pre-populated Seed Data	Phase 1	2 Days	2026-08-04	2026-08-05	1.1.2
1.1.4	Validation Rules & Business Logic	Phase 1	3 Days	2026-08-04	2026-08-06	1.1.2
1.1.5	UML Diagram & Milestone 1 Report	Phase 1	2 Days	2026-08-07	2026-08-08	1.1.3, 1.1.4
1.2	Milestone 2: JavaFX UI,	Phase 2	7 Days	2026-08-09	2026-08-15	1.1

Threads & Sockets						
1.2.1	Navigation Framework & CSS Shell	Phase 2	2 Days	2026-08-09	2026-08-10	1.1.4
1.2.2	Authentication Screens (Login/Register)	Phase 2	2 Days	2026-08-11	2026-08-12	1.2.1
1.2.3	Customer Dashboard & Menu/Order UI	Phase 2	3 Days	2026-08-11	2026-08-13	1.2.1
1.2.4	Staff & Admin Management Screens	Phase 2	3 Days	2026-08-11	2026-08-13	1.2.1
1.2.5	Multithreading & Socket Integration (Bonus)	Phase 2	2 Days	2026-08-13	2026-08-14	1.2.3, 1.2.4
1.2.6	Final Deliverables, Video & Git Clean-up	Phase 2	2 Days	2026-08-14	2026-08-15	1.2.5

3. Detailed Task Descriptions & Deliverables

- **1.1.1 Base Architecture & Shared Resources**

Establish initial Git repository, package hierarchy (model, controller, db, view), abstract base classes (User/Staff), and fundamental domain Enums (Role, TableStatus, OrderStatus).

- **1.1.2 Core Domain Entities & Interfaces**

Develop model classes for Customer, Admin, Waiter, Table, MenuItem, MenuCategory, Reservation, Order, OrderItem, and Invoice. Implement contracts like Payable and Manageable.

- **1.1.3 Data Store & Pre-populated Seed Data**

Implement RestaurantDatabase with static ArrayList storage for all entities. Pre-populate mock datasets for instant system testing upon startup.

- **1.1.4 Validation Rules & Business Logic**

Enforce validation logic: prevent double-booking, validate customer phone/password formats, enforce party size vs. table capacity, and perform billing calculations (taxes and fees).

- **1.1.5 UML Diagram & Milestone 1 Report**

Generate comprehensive UML Class Diagram detailing inheritance and relationships. Write design rationale section for the Milestone 1 report.

- **1.2.1 Navigation Framework & CSS Shell**

Set up main JavaFX stage, view switcher manager, and global CSS stylesheet for consistent UI theme across screens.

- **1.2.2 Authentication Screens**

Build FXML layouts and controllers for Login and Customer Registration with real-time field validation feedback.

- **1.2.3 Customer Dashboard & Menu/Order UI**

Implement interactive customer screens for viewing profile, browsing available tables with filters, making reservations, dynamic menu ordering, and checkout.

- **1.2.4 Staff & Admin Management Screens**

Implement Admin controls (CRUD on menu items and tables) and Waiter interface for managing active table seating and updating order statuses.

- **1.2.5 Multithreading & Socket Integration (Bonus)**

Add JavaFX Task/Service background loading and multi-threaded Socket server for real-time live order status synchronization between Waiter and Customer screens.

- **1.2.6 Final Deliverables, Video & Git Clean-up**

Compile test case matrix, record YouTube demo video, organize GitHub repository, and format final deliverable documentation.